

Table 9: Percentage of responses per dataset per model that our answer parser could not extract an answer for in the zero-shot direct answer setting. Prompt modifications were made to decrease these numbers. No model is above 15%.

Zero-shot Direct Answer Unparseable Answer Rate by Percentage														
dataset	Meta-Llama 2 7b	Mistral 7b	Meta-Llama 3.1 8b	Meta-Llama 3.1 70b	Gemma 2 9b	Phi-3 Small 8k	Qwen 2 7b	Qwen 2 72b	GPT-4o Mini	Gpt-4o	Claude 3 Haiku	Claude 3.5 Sonnet	Gemini 1.5 Flash	Gemini 1.5 Pro
CommonsenseQA	1.9	2.5	1.1	0.0	0.8	0.1	1.6	0.7	0.0	0.0	0.1	0.0	0.1	0.2
StrategyQA	0.0	1.9	0.1	0.0	11.7	0.5	4.9	2.7	0.0	0.0	0.0	0.0	0.0	0.2
SiQA	0.2	6.6	0.0	0.1	3.9	0.3	0.1	3.0	0.1	0.1	0.0	0.0	0.0	0.4
PiQA	0.4	6.0	0.0	0.1	3.3	2.1	0.0	5.5	0.2	0.0	0.1	0.0	0.1	0.9
Winogrande	0.0	3.0	0.0	0.1	2.1	0.2	5.1	0.4	0.0	0.0	0.0	0.0	0.0	3.6
Arc Easy	0.0	1.8	0.5	0.0	0.0	0.2	9.1	0.7	3.5	0.4	0.2	0.0	0.0	3.2
Arc Challenge	0.0	2.3	1.0	0.0	0.3	0.7	10.7	0.7	10.0	0.7	0.0	0.0	0.0	5.0
AGIEval LSAT LR	0.4	0.0	0.0	0.0	0.0	0.2	0.0	0.0	0.0	2.5	0.0	0.0	0.2	0.2
AGIEval LSAT AR	0.4	0.0	0.0	0.0	4.3	3.9	0.0	0.0	0.0	8.7	0.0	0.0	0.0	0.0
AGIEval LSAT RC	0.4	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	9.7	0.0	0.0	0.4	0.4
ContextHub Deductive L1	0.0	0.0	0.0	0.0	1.2	0.0	2.3	0.0	0.0	0.0	0.2	0.0	0.0	0.2
ContextHub Deductive L2	0.0	0.0	0.0	0.0	0.0	0.0	2.2	1.0	0.0	0.0	2.8	0.0	0.0	0.0
ContextHub Abductive L1	0.0	0.0	0.0	0.0	0.0	0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
ContextHub Abductive L2	0.0	0.0	0.0	0.0	0.0	0.1	1.5	0.2	0.0	0.0	0.8	0.0	0.0	0.0
MuSR Murder Mysteries	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
MuSR Team Allocations	0.0	0.0	0.0	0.0	3.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	8.4	0.4
MuSR Object Placements	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
MMLU	0.1	0.0	0.0	0.0	0.1	0.2	3.6	1.2	0.6	0.0	1.3	0.3	0.2	0.7
MMLU Pro	0.7	1.3	1.0	0.3	1.0	3.7	6.8	12.2	0.4	0.3	0.3	0.4	0.6	0.8
GPQA	1.3	7.1	0.0	0.0	8.7	12.7	5.4	15.2	0.0	0.0	1.6	0.0	0.0	0.7
MATH	0.6	6.9	0.3	0.2	0.1	0.1	3.5	3.0	0.8	0.0	0.3	0.0	0.4	0.6
GSM8k	0.2	4.1	2.5	0.0	2.7	0.0	1.7	0.2	0.0	0.0	12.7	5.5	0.0	0.0
BigGen Bench	4.6	0.3	0.9	0.1	0.5	1.0	1.3	1.0	1.3	0.0	0.0	0.1	0.4	0.3
GSM8k-Hard	4.8	7.6	2.0	0.4	0.2	0.2	3.2	1.1	0.1	0.5	5.2	0.5	0.2	0.0
MuSiQue	0.1	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.1
Folio	4.4	0.0	0.0	0.0	0.0	0.0	3.9	0.0	0.0	12.3	0.0	0.0	0.0	0.5
BigBench-Hard	0.0	0.0	0.0	7.4	0.0	0.2	0.0	0.0	0.0	0.3	0.0	4.5	0.2	12.8

Table 10: Percentage of responses per dataset per model that our answer parser could not extract an answer for in the zero-shot CoT setting. Prompt modifications were made to decrease these numbers. No model is above 15%.

Zero-shot CoT Unparseable Answer Rate by Percentage														
dataset	Meta-Llama 2 7b	Mistral 7b	Meta-Llama 3.1 8b	Meta-Llama 3.1 70b	Gemma 2 9b	Phi-3 Small 8k	Qwen 2 7b	Qwen 2 72b	GPT-4o Mini	Gpt-4o	Claude-3 Haiku	Claude-3.5 Sonnet	Gemini 1.5 Flash	Gemini 1.5 Pro
CommonsenseQA	2.9	1.3	8.6	0.0	0.6	0.1	0.0	0.0	1.6	0.0	0.2	0.3	2.4	2.6
StrategyQA	1.0	0.1	1.1	0.8	0.3	0.4	0.3	0.0	0.0	0.0	0.0	0.0	2.1	4.4
SiQA	0.8	1.8	0.3	0.1	1.6	0.0	0.1	0.1	0.0	0.0	0.3	0.1	3.5	4.0
PiQA	1.6	1.6	0.2	0.1	2.8	0.3	0.5	0.3	0.0	0.0	1.4	0.3	4.6	4.6
Winogrande	0.9	1.4	0.2	0.2	0.9	0.4	0.3	0.0	0.0	0.0	0.0	0.0	2.0	3.4
Arc Easy	0.2	0.4	0.2	0.0	0.5	1.6	1.6	0.0	0.5	0.0	0.0	0.0	0.4	0.5
Arc Challenge	0.0	0.7	0.0	0.0	0.0	0.0	0.7	0.0	0.0	0.0	0.0	0.0	0.7	0.7
AGIEval LSAT LR	3.3	2.2	0.0	0.0	1.2	0.0	2.0	0.0	0.0	0.0	0.0	0.0	0.8	0.2
AGIEval LSAT AR	4.8	7.0	6.1	2.2	5.7	5.2	4.3	0.4	1.3	1.3	0.0	0.4	4.8	1.7
AGIEval LSAT RC	7.1	1.1	0.0	0.0	0.7	3.0	6.7	0.0	0.0	0.0	0.0	0.0	0.0	0.4
ContextHub Deductive L1	0.7	1.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.3
ContextHub Deductive L2	0.2	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.5	0.4
ContextHub Abductive L1	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
ContextHub Abductive L2	0.0	0.2	0.1	0.0	0.0	0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.5	0.4
MuSR Murder Mysteries	0.0	0.4	0.0	0.0	0.0	11.6	0.4	0.0	0.0	0.0	0.0	0.0	6.8	3.6
MuSR Team Allocations	5.2	3.2	0.8	0.0	0.8	0.4	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0
MuSR Object Placements	0.0	1.6	0.0	0.0	0.4	0.8	0.0	0.0	0.0	0.0	0.0	0.0	2.0	0.4
MMLU	1.9	0.6	1.0	0.2	1.5	1.0	0.4	0.2	0.0	0.1	0.0	0.1	3.1	3.2
MMLU Pro	4.4	5.4	13.1	3.3	12.5	3.6	5.4	2.0	2.4	1.9	0.4	0.4	5.0	4.4
GPQA	4.5	10.3	9.4	1.6	8.5	1.8	3.8	0.7	0.0	0.0	0.0	0.0	11.8	15.0
MATH	1.6	5.5	8.2	2.5	2.3	1.6	3.0	0.4	0.4	0.5	0.9	0.0	1.7	1.0
GSM8k	1.7	1.4	0.7	10.5	0.4	0.6	0.4	0.0	0.0	0.0	0.3	0.0	0.1	0.1
BigGen Bench	5.0	0.4	0.5	0.1	0.5	0.4	0.3	9.5	0.0	0.0	0.0	0.1	0.4	0.1
GSM8k-Hard	2.1	8.7	10.2	4.5	10.7	3.2	3.5	1.0	0.8	0.5	3.0	1.8	0.4	2.7
MuSiQue	1.4	0.0	8.3	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.7	3.1
Folio	0.0	0.0	1.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	2.0	1.5
BigBench-Hard	3.8	5.4	1.8	0.4	1.3	0.1	0.4	0.3	0.0	0.0	0.0	0.0	1.2	0.9

Table 11: Percentage of responses per dataset per model that our answer parser could not extract an answer for in the few-shot direct answer setting. Prompt modifications were made to decrease these numbers. No model is above 15%.

Few-shot Direct Answer Unparseable Answer Rate by Percentage										
dataset	Meta-Llama 2 7b	Mistral 7b	Meta-Llama 3.1 8b	Meta-Llama 3.1 70b	Gemma 2 9b	Phi-3 Small 8k	Qwen 2 7b	Qwen 2 72b	GPT-4o Mini	Gemini 1.5 Flash
CommonsenseQA	0.0	0.1	0.2	0.0	1.3	0.9	9.9	1.3	0.0	0.6
AGIEval LSAT LR	6.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2
AGIEval LSAT AR	2.6	0.0	0.0	0.0	3.5	5.2	0.0	0.0	0.0	0.0
AGIEval LSAT RC	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
ContextHub Deductive L1	0.0	2.8	0.0	0.0	0.0	10.7	0.3	0.0	0.0	0.0
ContextHub Deductive L2	0.0	0.1	0.0	0.0	0.0	0.3	0.2	0.0	0.0	0.0
ContextHub Abductive L1	0.0	2.8	0.0	0.0	0.0	0.6	0.0	0.0	0.0	0.0
ContextHub Abductive L2	0.0	2.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
MuSR Murder Mysteries	-1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.4
MuSR Team Allocations	-1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
MuSR Object Placements	-1.0	0.0	0.0	0.0	0.4	1.2	0.0	0.0	0.0	0.0
MMLU	4.2	0.2	0.0	0.0	0.1	0.0	0.4	0.1	0.0	0.2
MMLU Pro	5.1	1.2	2.4	0.3	1.0	9.1	0.5	2.6	0.4	0.5
GPQA	-1.0	1.3	0.0	0.0	3.6	7.4	13.4	1.1	0.0	0.0
MATH	0.3	5.9	0.3	0.2	0.1	0.1	1.6	2.2	0.0	0.3
GSM8k	0.1	0.1	0.5	0.0	0.1	2.2	0.0	0.2	0.0	0.0

Table 12: Percentage of responses per dataset per model that our answer parser could not extract an answer for in the few-shot CoT setting. Prompt modifications were made to decrease these numbers. No model is above 15%.

Few-shot CoT Unparseable Answer Rate by Percentage										
dataset	Meta-Llama 2 7b	Mistral 7b	Meta-Llama 3.1 8b	Meta-Llama 3.1 70b	Gemma 2 9b	Phi-3 Small 8k	Qwen 2 7b	Qwen 2 72b	GPT-4o Mini	Gemini 1.5 Flash
CommonsenseQA	0.7	0.9	1.8	0.1	0.2	0.1	0.0	0.0	0.0	3.4
AGIEval LSAT LR	0.6	0.8	0.4	0.0	1.4	3.1	0.8	0.0	0.0	0.6
AGIEval LSAT AR	2.2	9.1	3.9	0.9	11.7	3.0	3.5	1.7	0.0	1.3
AGIEval LSAT RC	7.8	5.9	0.0	0.0	1.9	9.3	2.6	0.0	0.0	2.2
ContextHub Deductive L1	0.2	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.3
ContextHub Deductive L2	0.9	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.3
ContextHub Abductive L1	0.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
ContextHub Abductive L2	3.1	0.0	5.3	0.1	0.0	0.2	0.0	0.0	0.0	0.7
MuSR Murder Mysteries	-1.0	1.2	0.0	0.0	0.4	0.8	0.0	0.0	0.0	14.0
MuSR Team Allocations	-1.0	2.4	0.0	0.0	0.0	0.0	0.8	0.0	0.0	0.4
MuSR Object Placements	-1.0	0.4	0.0	0.0	1.2	0.4	0.0	0.0	0.0	0.0
MMLU	0.6	0.8	1.1	0.2	1.5	0.7	0.3	0.2	0.2	2.5
MMLU Pro	0.6	1.9	8.5	2.1	14.1	1.8	1.9	0.8	1.1	3.9
GPQA	-1.0	12.1	10.3	0.9	12.9	6.0	5.6	3.3	0.0	13.6
MATH	1.5	6.8	8.2	2.4	11.1	2.6	2.9	1.1	0.5	1.8
GSM8k	0.8	1.3	1.0	0.1	0.5	0.5	0.1	0.0	0.1	0.1

Table 13: Average character index of where the answer span begins in a generated response for each dataset and model pair for the zero-shot direct answer setting. We use these numbers as a proxy for the model following instructions (i.e. generating reasoning before an answer). Prompt modifications were made to ensure CoT prompts resulted in longer generations and direct answer prompts led to short generations.

Zero-shot Direct Answer Span Location By Character Index													
dataset	Meta-Llama 2.7b	Mistral 7b	Meta-Llama 3.1 8b	Meta-Llama 3.1 70b	Gemma 2 9b	Phi-3 Small 8k	Qwen 2.7b	Qwen 2.72b	GPT-4o Mini	GPT-4o	Claude-3 Haiku	Claude-3.5 Sonnet	Gemini 1.5 Flash
CommonsenseQA	9	8	27	8	8	8	10	8	8	10	7	7	8
StrategyQA	44	45	27	44	44	44	46	44	8	8	42	41	8
SiQA	8	8	8	8	8	8	29	8	8	8	6	6	8
PiQA	7	8	8	8	8	8	25	8	8	8	4	5	8
Winogrande	8	9	8	8	8	8	9	8	8	9	5	4	8
Arc Easy	9	8	8	8	8	8	9	8	8	8	7	7	8
Arc Challenge	8	8	8	8	8	8	9	8	8	8	7	7	8
AGIEval LSAT LR	25	24	24	24	24	24	25	24	43	21	25	25	26
AGIEval LSAT AR	25	24	24	24	24	24	26	24	48	23	25	25	26
AGIEval LSAT RC	25	24	24	24	24	24	25	24	31	18	25	25	26
ContextHub Deductive L1	19	19	19	19	20	19	19	19	19	19	20	20	19
ContextHub Deductive L2	19	19	19	19	19	19	19	19	19	19	20	20	19
ContextHub Abductive L1	19	19	19	19	20	19	19	19	19	19	20	20	19
ContextHub Abductive L2	19	19	19	19	20	19	19	19	19	19	20	20	19
MuSR Murder Mysteries	8	8	27	8	8	8	8	8	8	8	6	4	8
MuSR Team Allocations	27	22	19	19	27	23	26	22	8	8	30	20	8
MuSR Object Placements	8	8	27	8	8	8	8	8	8	8	7	6	8
MMLU	19	18	19	19	20	18	18	18	19	19	19	19	19
MMLU Pro	20	19	38	19	21	19	19	20	19	19	20	20	19
GPQA	19	19	19	19	21	19	19	19	19	19	20	20	19
MATH	30	31	28	28	28	30	30	33	28	28	31	29	28
GSM8k	22	29	30	28	28	37	24	28	28	28	29	28	28
GSM8k-Hard	9	57	11	21	9	13	40	20	7	8	8	8	8
Folio	39	8	8	8	8	8	31	13	8	16	5	6	8
BigBench-Hard	39	22	25	21	26	32	29	26	28	19	28	28	10

Table 14: Average character index of where the answer span begins in a generated response for each dataset and model pair for the zero-shot CoT setting. We use these numbers as a proxy for the model following instructions (i.e. generating reasoning before an answer). Prompt modifications were made to ensure CoT prompts resulted in longer generations and direct answer prompts led to short generations.

Zero-shot CoT Answer Span Location By Character Index													
dataset	Meta-Llama 2.7b	Mistral 7b	Meta-Llama 3.1 8b	Meta-Llama 3.1 70b	Gemma 2 9b	Phi-3 Small 8k	Qwen 2.7b	Qwen 2.72b	GPT-4o Mini	GPT-4o	Claude-3 Haiku	Claude-3.5 Sonnet	Gemini 1.5 Flash
CommonsenseQA	441	564	845	1237	236	466	577	341	899	1086	626	1103	214
StrategyQA	726	434	996	1131	267	460	363	358	692	1033	754	1158	256
SiQA	569	423	841	965	235	528	472	420	847	1094	602	1016	196
PiQA	699	455	869	914	207	532	447	364	683	935	578	1092	200
Winogrande	377	324	645	694	187	326	391	298	634	750	408	889	200
Arc Easy	684	581	1154	1319	367	610	534	355	990	1239	789	1222	340
Arc Challenge	763	644	1178	1316	422	596	571	387	1020	1269	828	1240	372
AGIEval LSAT LR	2053	1324	1163	1675	524	689	1560	768	949	998	1561	728	906
AGIEval LSAT AR	1377	1791	1422	2182	712	1027	1819	1264	1230	1151	1202	849	817
AGIEval LSAT RC	1977	1032	1103	1575	779	590	1170	660	973	1079	1628	786	703
ContextHub Deductive L1	694	368	759	711	383	327	539	402	540	580	542	556	320
ContextHub Deductive L2	842	472	1095	990	614	442	789	585	840	758	777	655	515
ContextHub Abductive L1	577	461	747	879	464	440	754	638	788	879	683	594	368
ContextHub Abductive L2	861	600	1270	1229	686	571	976	856	1115	1113	894	894	601
MuSR Murder Mysteries	495	1592	1958	1847	1210	1246	1241	1718	1961	1965	1671	1759	1349
MuSR Team Allocations	1212	1845	2294	2310	1513	1433	2021	2213	2562	2698	1479	1856	1596
MuSR Object Placements	917	625	1354	1266	695	641	904	819	1593	1536	1210	1455	616
MMLU	834	512	663	622	503	277	497	407	400	461	447	409	630
MMLU Pro	1371	513	788	716	640	518	954	699	926	940	590	653	660
GPQA	1034	778	917	901	806	500	1018	628	541	666	486	472	981
MATH	742	1118	1222	1179	748	670	1189	1145	1125	1153	677	675	679
GSM8k	572	637	834	719	453	521	709	645	1048	1035	708	680	541
GSM8k-Hard	916	939	1027	1069	555	766	1083	1053	1350	1266	594	815	605
Folio	724	765	1479	1379	733	668	919	488	1285	1583	907	1194	934
BigBench-Hard	596	230	876	861	429	349	315	443	877	973	545	863	455